

## Lab 5 - Texture synthesis Image Processing

### References:

- [http://www.ipol.im/pub/art/2011/ggm\\_rpn/](http://www.ipol.im/pub/art/2011/ggm_rpn/)
- <http://www.ipol.im/pub/art/2013/59/>
- Class slides

### 1 Modulus and phase of the DFT of an image



Figure 1: Left: image  $u_1$  (roma.png). Right: image  $u_2$  (texture.png).

Given two images  $u_1$  and  $u_2$  (Fig. 1) and their discrete Fourier transform  $\hat{u}_1$  and  $\hat{u}_2$  generate two new images  $v_1$  and  $v_2$  whose discrete Fourier transform are defined as follows:

$$\hat{v}_1 = |\hat{u}_1|e^{i\arg(\hat{u}_2)} \quad (1)$$

and

$$\hat{v}_2 = |\hat{u}_2|e^{i\arg(\hat{u}_1)}. \quad (2)$$

Here  $|\hat{u}_i|$ ,  $i = 1, 2$  is the modulus of the discrete Fourier transform of  $u_i$ ,  $i = 1, 2$  and  $\arg(\hat{u}_i)$ ,  $i = 1, 2$  is the phase of the discrete Fourier transform of  $u_i$ ,  $i = 1, 2$ .

- Create a Python code to compute  $v$  and  $v$ .
- Comment and explain the results you obtained.

## 2 Random Phase Noise (RPN)

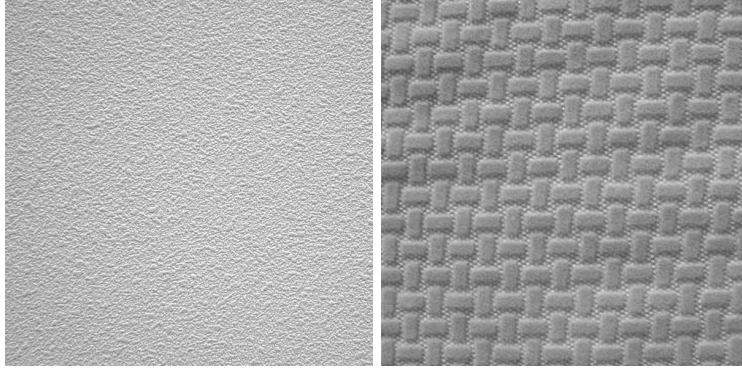


Figure 2: Left: input image textil-1-gray.png. Right: input image textil-2-gray.png

The RPN texture synthesis method consist in synthesizing a texture image by randomizing its Fourier phase. That is, given an input texture  $u$ , the synthesized image  $v$  is defined as:

$$\hat{v}(s, t) = |\hat{u}(s, t)|e^{i\phi(s, t)}, \quad (s, t) \in \hat{\Omega} \quad (3)$$

where  $\phi(s, t)$  is a random phase. The algorithm is as follows:

1. Generate a random phase  $\phi$  (hint: the phase of the DFT of a Gaussian white noise is a random phase)
2. Compute the DFT of the input texture  $\hat{u}$
3. Compute  $\hat{v}$  as defined in (3)
4. Compute the texture  $v$  as the inverse DFT of  $\hat{v}$

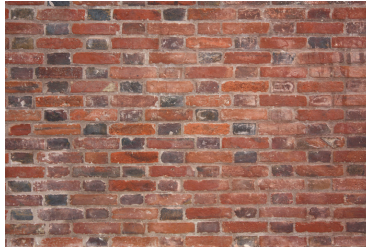


Figure 3: Input image bricks.png.

- a) Create a python function that performs the previous algorithm to synthesize the textures: `textil-1-gray.png` and `textil-2-gray.png` (see Figure 2).
- b) Comment the results.
- c) Extend this algorithm to colour images and synthesize texture `bricks.png` (see Figure 7). (Be careful with the coherence of the colour channels, new colours should not appear.)
- d) Comment the difference between the grayscale version and the colour version.

### 3 Efros and Leung's method

Given the input textures  $u_1$  and  $u_2$  in Figure 4 and the synthesized textures generated with the Efros and Leung's method in Figures 5 and 6 find the correct patch size used in each synthesis. That is indicate for  $v_1, v_2, v_3, v_4, v_5$  and  $v_6$  in Figures 5 and 6 what patch size was used. The options are:  $5 \times 5$ ,  $9 \times 9$  and  $33 \times 33$ . Justify your choice in each case.

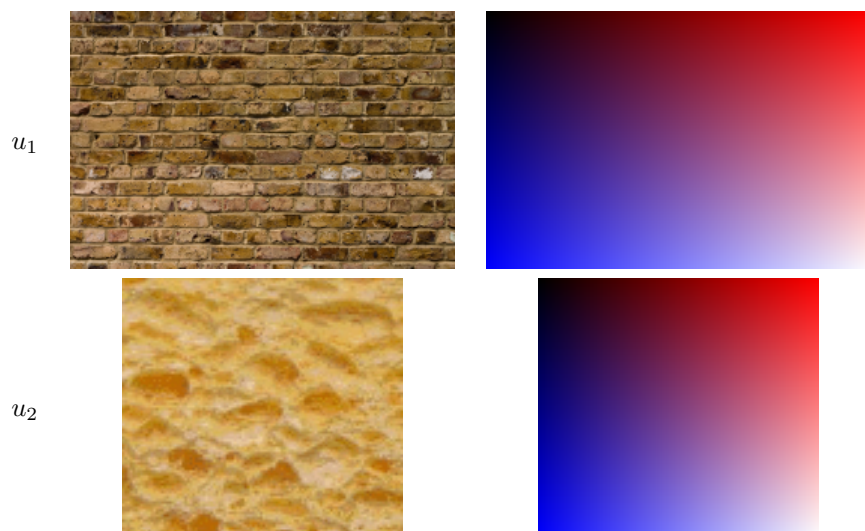


Figure 4: First row: on the left input image  $u_1$  (bricks-2.png) and on the right its corresponding color map. Second row: on the left input image  $u_2$  (texture-2.png) and on the right its corresponding color map.

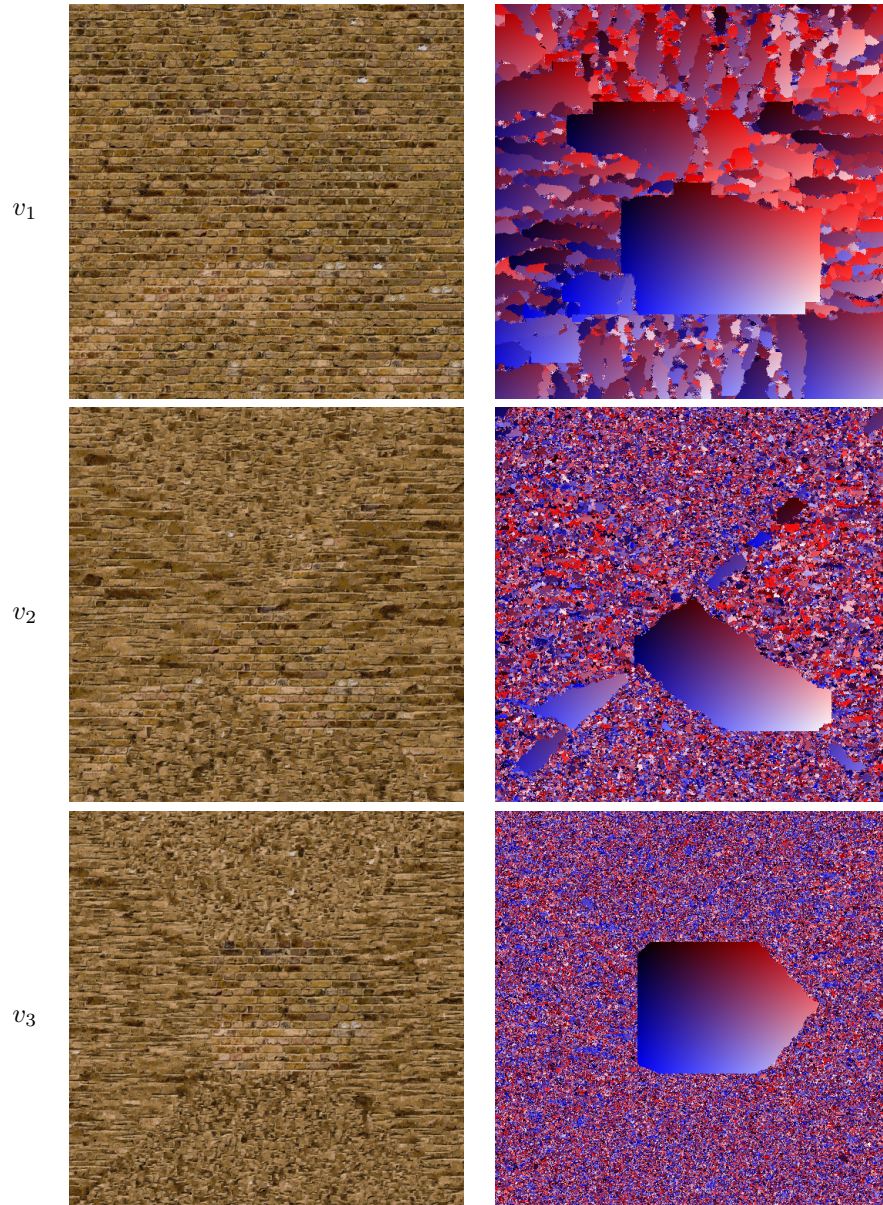


Figure 5: Each row has a texture generated with the Efros and Leung's method and its corresponding copy map. Each row was generated with a different patch size.



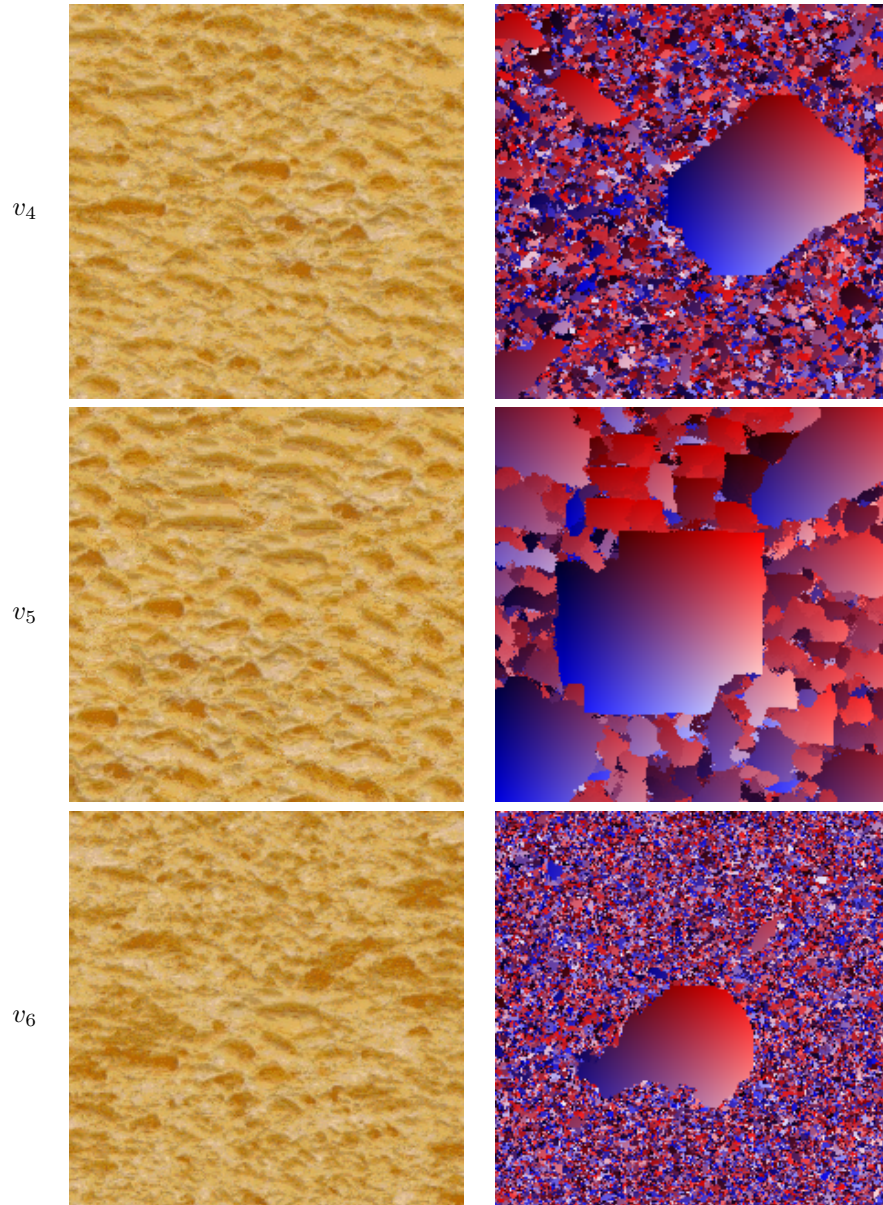


Figure 6: Each row has a texture generated with the Efros and Leung's method and its corresponding copy map. Each row was generated with a different patch size.

## 4 Optional



Figure 7: Input image clouds.png.

- a) Synthesize image clouds.png (see Figure 7) using the RPN method implemented in exercise 2 part c). Comment the result.
- b) Generate a symmetric version  $w$  of cloud.png of size  $2M \times 2N$ , where  $M \times N$  is the size of clouds.png, which follows the following scheme:

|   |   |
|---|---|
| F | T |
| E | A |

- c) Synthesize the image  $w$  using the RPN method implemented in exercise 2 part c).
- d) Extract the upper left size of the synthesized image of size  $M \times N$ . Compare to the result in a).
- e) Explain the difference between both results.